INCH-POUND

MIL-DTL-17/196C
w/AMENDMENT 3
23 February 2024
SUPERSEDING
MIL-DTL-17/196C
w/ AMENDMENT 2
18 May 2018

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/196-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

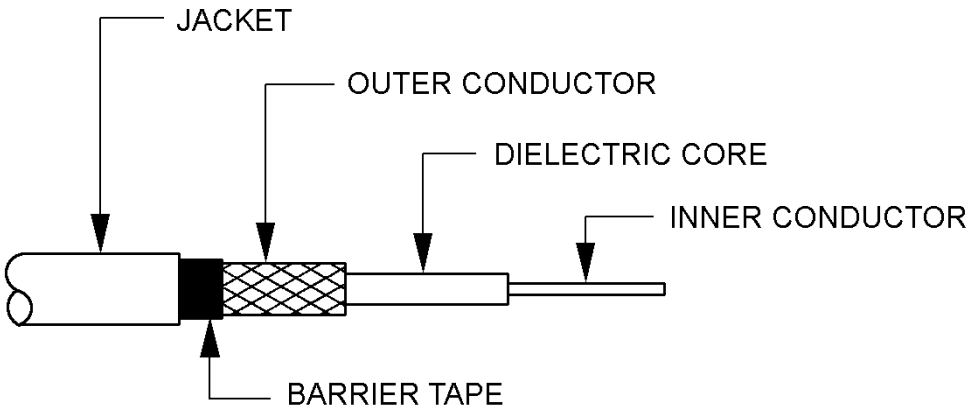


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of copper-covered steel wire, each strand .0063 inch diameter. Overall diameter: .0189 inch ± .0010.
Dielectric core	Type A-1: Solid polyethylene. Diameter: .060 inch ± .003.
Outer conductor	Single braid of AWG No. 38, tinned copper wire. Diameter: .083 inch maximum. Coverage: 85.6% nominal Carriers: 16 Ends: 4 Picks/inch: 16.3 ± 10%
Barrier tape	A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape will be applied with a 50% lap, aluminum face toward the outer conductor. Diameter: .093 inch maximum.
Jacket	Cross-linked polyolefin. Diameter: .110 inch ± .005. Jacket thickness: .008 inch minimum.

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ENGINEERING INFORMATION:

Continuous working voltage: 1,100 V rms, maximum.
Operating frequency: 1 GHz, maximum.
Velocity of propagation: 65.9 percent, nominal.
Power ratings: See figure 2.
Operating temperature range: -30°C to +85°C.
Inner conductor properties:
DC resistance (maximum at +20°C): 9.67 ohms per 100 feet.
Elongation: 1 percent, minimum.
Tensile strength: 110 $klb_f/inch^2$, minimum.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.
Environmental and mechanical:
Visual and mechanical examination: Applicable.
Diameter: See figure 1 and table I.
Out-of-roundness: See figure 1 and table I.
Eccentricity: 10 percent maximum.
Adhesion of conductors: Inner conductor to core: 4 pounds, minimum; 12 pounds, maximum.
Aging stability: +98°C ± 2°C.
Stress crack resistance: Not applicable.
Outer conductor integrity: Not applicable.
Cold bend: -30°C ± 2°C.
Dimensional stability: +85°C ± 2°C.
Inner conductor from core: .062 inch, maximum.
Inner conductor from jacket: .125 inch, maximum.
Bendability: Not applicable.
Flammability: Not applicable.
Weight: 0.88 pounds per 100 feet maximum.

Electrical:

Continuity: Applicable.
Spark test: 2,000 V rms, minimum.
Voltage withstanding: 4,500 V rms, minimum.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,500 V rms, minimum.
Characteristic impedance: 50 ± 2 ohms.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 32.2 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.
Contamination: Not applicable.
Inductance: Not applicable.
Marking: Not applicable.
Marking durability: Not applicable.
Workmanship: Not applicable.

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Flame propagation: Applicable.
Acid gas generation: 2.0 percent, maximum.
Halogen content: 0.2 percent, maximum.
Immersion test:
Tensile strength, percent of unaged minimum: 50
Elongation, percent of unaged minimum: 50.
Smoke index: 25 maximum.
Toxicity index: 5 maximum.
Durometer hardness: (Type A) 80 minimum.
Weathering: Jacket material shall be tested in accordance with Measurement 3306 of MIL-STD-1678-3.
Abrasion resistance: 75 cycles minimum (jacket only).
Tear strength: 35 pounds per inch minimum.
Heat distortion: 30 percent maximum distortion.
Physical tests on unaged jacket:
Tensile strength: 1,300 psi, minimum.
Elongation, 160 percent, minimum.
Physical tests on aged jacket:
Air oven:
Tensile strength, percent minimum: 60
Elongation, percent minimum: 60
Hot oil immersion:
Tensile strength, percent minimum: 50
Elongation, percent minimum: 50
Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Part or Identifying Number (PIN): M17/196-00001.

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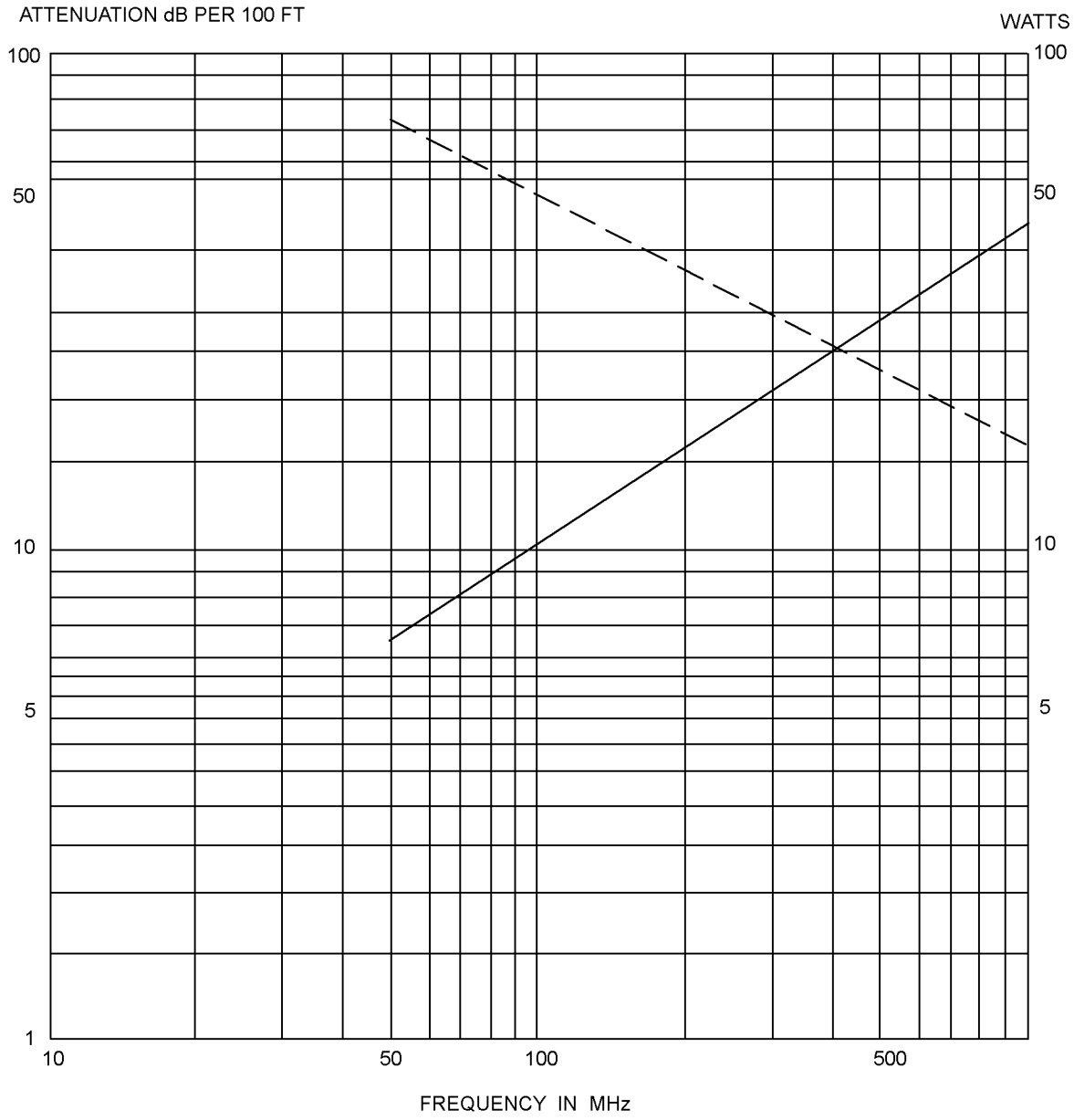


FIGURE 2. Power rating and attenuation.

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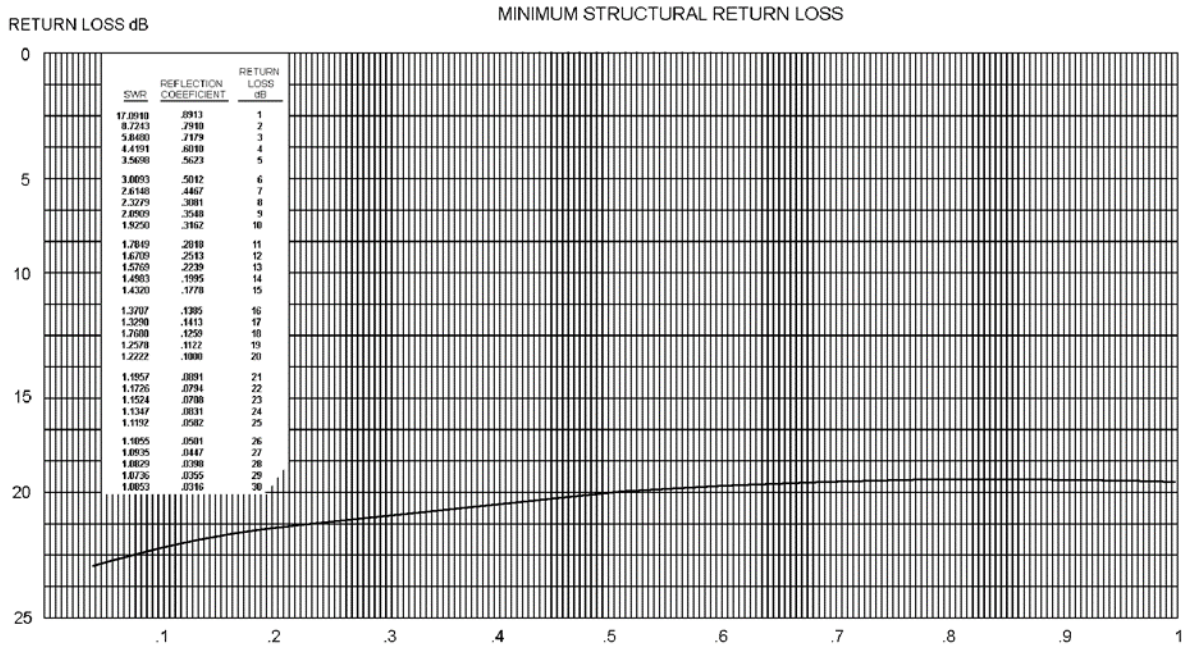


FIGURE 3. Structural return loss.

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TABLE II. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness of jacket measurements	3.6.2	4.8.1.2	Entire sample	Per Table VIII	A
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	N/A				
	Marking Durability	N/A				
	Workmanship	N/A				
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
II	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	3.7.8	4.8.9	2	Per Table VIII	A
	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B

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TABLE II. Qualification and Group A, B, and C Table - Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Aging Stability	3.7.15				C
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Weight	3.9	4.8.24	1	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Flame propagation	3.7.23	4.8.25	1		
	Acid gas generation	3.7.24	4.8.26	1		
	Halogen content	3.7.25	4.8.27	1		
	Immersion	3.7.26	4.8.28	1		
	Smoke index	3.7.27	4.8.29	1		
	Toxicity index	3.7.28	4.8.30	1		
	Durometer hardness	3.7.29	4.8.31	1		
	Weathering	3.7.30	4.8.32	1		
	Abrasion resistance	3.7.31	4.8.33	5		
	Tear strength	3.7.32	4.8.34	6	Per Table VIII	B
	Heat distortion	3.7.33	4.8.35	1	Per Table VIII	C
	Physicals	3.7.34	4.8.36	1	Per Table VIII	C
	Tensile strength and elongation	3.7.35	4.8.37	1	Per Table VIII	C

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Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17 this document references the following:

MIL-STD-1678-3

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activities:
Army – AR, AT, CR4
Navy – AS, MC, OS
Air Force – 19

(Project 6145-2024-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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